Complex Analysis Homework 4

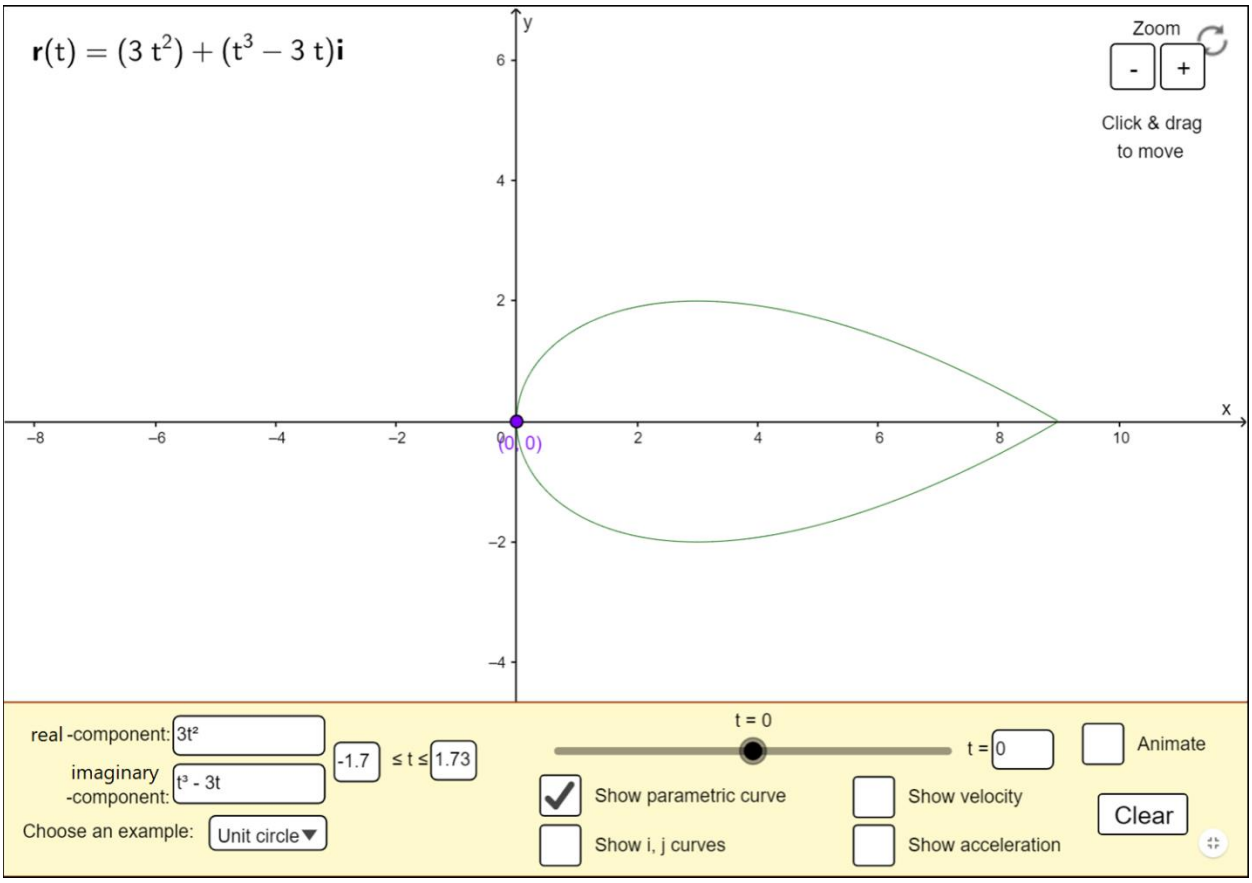
1. Without solving the following contour integrals, show that

(a) $\left|\int_0^{1+i} z^2 dz\right| \le 2\sqrt{2},$

(b) $\left|\int_{|z-1|=1} \frac{e^z}{z+1} dz\right| \le 2\pi e^2,$ in counterclockwise direction

(c) $\left|\int_{|z|=R} \frac{z^n}{z^m - 1} dz\right| \underset{\text{strict}}{<} \frac{2\pi R^{n+1}}{R^m - 1},$ $R > 1, m \ge 1, n \ge 0,$ in counterclockwise direction

(d) $\left|\int_C \frac{Arg(z-i)}{z-1} dz\right| \le 6\sqrt{3}\pi$ where $C : z(t) = 3t^2 + (t^3 - 3t)i, -\sqrt{3} \le t \le \sqrt{3}.$



2. Let γ be the circle $\{|z| = 1\}$, with the usual counterclockwise orientation. Evaluate the following integrals, for $m = 0, \pm 1, \pm 2, \ldots$

(a) $\int_\gamma z^m |dz|$

(b) $\int_\gamma \overline{z}^m dz.$

3. Let γ be the circle $\{|z| = R\}$, with the usual counterclockwise orientation. Evaluate the following integrals, for $m = 0, \pm 1, \pm 2, \ldots$

(a) $\int_\gamma \left|z^m\right| dz$

(b) $\int_\gamma \left|z^m\right| |dz|$

(c) $\int_\gamma \overline{z}^m dz.$

4. (a) Let γ be the circle $\{|z| = 1\}$, with the usual counterclockwise orientation. Using the principal branch, evaluate $\int_\gamma z^{-1+i} dz.$

(b) Let γ be the boundary of the triangle $\{0 \le y \le 1 - x, 0 \le x \le 1\}$, with the usual counterclockwise orientation. Evaluate the following integrals

(i) $\int_\gamma \operatorname{Re} z dz$

(ii) $\int_\gamma \operatorname{Im} z dz$

5. By evaluating the contour integral $\int_{|z|=1} \left(z + \frac{1}{z}\right)^{2n} \frac{dz}{z}$ where $|z| = 1$ in counterclockwise direction, show that $\int_0^{2\pi} \cos^{2n} \theta d\theta = 2\pi \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots (2n)}$